Gas 10-Problem Set: Worked Solutions

Each solution follows the 5-step routine: **table → convert → pick equation → predict → solve/check**. Red = common mistake. R = 0.08206 L·atm/(mol·K), STP = 273.15 K and 1 atm (22.414 L/mol).

1. `P₁=755 mmHg V₁=5.00 L T₁=298.15 K | P₂=380. mmHg V₂=? T₂=258.15 K`, n constant.
$V_2 = V_1 \times (P_1/P_2) \times (T_2/T_1) = 5.00 \times (755/380.) \times (258.15/298.15) = 5.00 \times 1.987 \times 0.8658 =$ **8.60 L**.
Pressure dropped by about half (×1.99 for V), but T dropped only 13% (×0.866). The pressure effect wins, so the balloon expands. **Trick: write each change as a ratio factor and ask whether it should make V bigger or smaller.**

---

2. n = 4.50/44.009 = 0.1023 mol; T = 310.15 K. P = nRT/V = (0.1023)(0.08206)(310.15)/2.00 = **1.30 atm**.

---

3. Empirical: C 85.6/12.011 = 7.127, H 14.4/1.008 = 14.29 → 1 : 2.00 → CH_2 (14.027 g/mol).
`m=0.802 g, V=0.354 L, T=373.15 K, P=752/760=0.9895 atm`
$\mathcal{M}$ = mRT/(PV) = (0.802)(0.08206)(373.15)/[(0.9895)(0.354)] = **70.1 g/mol**. k = 70.1/14.027 = 5.00 → **C_5H_{10}**.
Exam combo: % composition → empirical formula, gas data → $\mathcal{M}$, then k = $\mathcal{M}/\mathcal{M}_{emp}$.

---

4. n(He) = 2.00/4.0026 = 0.4997 mol; n(O_2) = 8.00/31.998 = 0.2500 mol. RT/V = (0.08206 × 298.15)/5.00 = 4.893 atm/mol.
P(He) = 0.4997 × 4.893 = **2.45 atm**; P(O_2) = 0.2500 × 4.893 = **1.22 atm**; P_tot = **3.67 atm**. χ(He) = 0.4997/0.7497 = **0.667** (check: 0.667 × 3.67 = 2.45 ✓).
The partial pressure depends on MOLES, not grams: 8 g O_2 gives less pressure than 2 g He.

---

5. P(O_2) = 758 − 19.8 = 738.2 mm-Hg = 0.9713 atm; T = 295.15 K; V = 0.512 L.
n(O_2) = (0.9713)(0.512)/(0.08206 × 295.15) = 0.02053 mol → × (2 $KClO_3$/3 O_2) = 0.01369 mol → × 122.55 = **1.68 g $KClO_3$**.
Two traps: forgetting to subtract the water vapor, and forgetting the 2:3 ratio.

---

6. n(N_2) = PV/RT = (1.10)(65.0)/(0.08206 × 298.15) = 2.922 mol → × (2/3) = 1.948 mol NaN_3 × 65.01 = **127 g NaN_3**.
Pattern: gas → mol via PV=nRT, then mole ratio, then grams.

---

7. Same T and P → **liters behave like moles**. 6.00 L N_2 would need 18.0 L H_2, but only 15.0 L is available → **H_2 limiting**. NH_3 = 15.0 × (2/3) = **10.0 L**. N_2 used = 15.0/3 = 5.00 L → **1.00 L N_2 left over**.

---

8. n(Al) = 2.00/26.982 = 0.07412 mol (÷2 = 0.0371). n(Cl_2) = (1.00)(1.50)/(0.08206 × 298.15) = 0.06131 mol (÷3 = 0.0204) → **Cl_2 is limiting**.
$AlCl_3$ = 0.06131 × (2/3) = 0.04087 mol × 133.33 = **5.45 g $AlCl_3$**.

---

9. (a) rate(H_2)/rate(O_2) = √(31.998/2.016) = **3.98× faster**.
(b) rate(X)/rate(He) = √($\mathcal{M}$_He/$\mathcal{M}$_X) → 0.354 = √(4.0026/$\mathcal{M}$) → $\mathcal{M}$ = 4.0026/0.354² = **31.9 g/mol → O_2**.
(c) Time is the inverse of rate: t_X/t_N_2 = √($\mathcal{M}$_X/$\mathcal{M}$_N_2) → $\mathcal{M}$ = 28.014 × (191/105)² = **92.7 g/mol**.
Heavier gas = slower rate = LONGER time. Square both sides to solve for $\mathcal{M}$.

---

10. u_rms = √(3RT/$\mathcal{M}$), R = 8.314 J/(mol·K), T = 298.15 K.
He: √(3 × 8.314 × 298.15/0.0040026) = **1.36 × 10³ m/s**; Xe: √(3 × 8.314 × 298.15/0.13129) = **238 m/s**.
(b) Equal speeds mean equal T/$\mathcal{M}$: T(O_2) = 298.15 × (31.998/4.0026) = **2.38 × 10³ K**.
$\mathcal{M}$ in kg/mol (4.0026 g → 0.0040026 kg). With g/mol you'd get an answer √1000 ≈ 31.6× too small.